

JEE MAINS & IIT JEE ADVANCED

MATHEMATICS

PARABOLA

DIBY (DO IT BY YOURSELF)

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DIBY-01

Equation of the parabola whose axis is parallel to Y-axis and which passes through the points $(1, 0)$, $(0, 0)$ and $(-2, 4)$ is

(A) $2x^2 + 2x = 4y$

(B) $2x^2 - 2x = 3y$

(C) $2x^2 + 2x = y$

(D) $2x^2 - 2x = y$

[Ans. (B)]

DIBY-02

Find the equation of the parabola whose focus is $(5, 3)$ and directrix is the line $3x - 4y + 1 = 0$.

[Ans. $16x^2 + 9y^2 + 24xy - 256x - 142y + 849 = 0$]

DIBY-03

Find the equation of the parabola is focus is at $(-6, -6)$ and vertex is at $(-2, 2)$.

[Ans. $4x^2 + y^2 - 4xy + 104x + 148y - 124 = 0$]



DIBY-04

Find the vertex, focus, axis, directrix and latusrectum of the parabola

$$4y^2 + 12x - 20y + 67 = 0$$

$$[\text{Ans. } \left(\frac{-7}{2}, \frac{-7}{2}\right); \left(\frac{-17}{2}, \frac{5}{2}\right); y = \frac{5}{2}; x + \frac{11}{4} = 0; 3]$$

DIBY-05

Find the name of the conic represented by $\sqrt{\left(\frac{x}{a}\right)} + \sqrt{\left(\frac{y}{b}\right)} = 1$

[Ans. Parabola]

DIBY-06

Determine the name of the curve described parametrically by the equations

$$x = t^2 + t + 1, y = t^2 - t + 1.$$

[Ans. Parabola]



DIBY-07

If the vertex of a parabola is $(2, 2)$ and the co-ordinates of its two extremities of the latus rectum are $(-2, 0)$ and $(6, 0)$, then find the equation of the parabola.

$$[\text{Ans. } (x - 2)^2 = -8(y - 2)]$$

DIBY-08

Find the length of the latus rectum of the parabola $169\{(x - 1)^2 + (y - 3)^2\} = (5x - 12y + 17)^2$.

$$[\text{Ans. } 28/13]$$

DIBY-09

If the line $x - 1 = 0$ is the directrix of the parabola $y^2 - kx + 8 = 0$, then find the value of k .

$$[\text{Ans. } -8, 4]$$

DIBY-10

Find the latus rectum of a parabola, whose focal chord PSQ is such that $SP = 3$ and $SQ = 2$.

[Ans. 24/5]

DIBY-11

The axis of a parabola is the line $y = x$ and its vertex and focus are in the first quadrant at distances $\sqrt{2}$ and $2\sqrt{2}$ units from the origin, respectively. If the point $(1, k)$ lies on the parabola, then a possible value of k is

[JEE Mains-2025]

- (A) 4 (B) 9 (C) 3 (D) 8

[Ans. (B)]

DIBY-12

Let P be the parabola, whose focus is $(-2, 1)$ and directrix is $2x + y + 2 = 0$. Then the sum of ordinates of the points on P , whose abscissa is -2 , is

[JEE Mains-2025]

- (A) $\frac{3}{2}$ (B) $\frac{5}{2}$ (C) $\frac{1}{4}$ (D) $\frac{3}{4}$

[Ans. (A)]

DIBY-13

The focus of the parabola $y^2 = 4x + 16$ is the centre of the circle C of radius 5. If the values of λ , for which C passes through the point of intersection of the lines $3x - y = 0$ and $x + \lambda y = 4$, are λ_1 and λ_2 , $\lambda_1 < \lambda_2$, then $12\lambda_1 + 29\lambda_2$ is equal to

[JEE Mains-2025]

[Ans. : 15]

DIBY-14

If the equation of the parabola with vertex $V\left(\frac{3}{2}, 3\right)$ and the directrix $x + 2y = 0$ is

$\alpha x^2 + \beta y^2 - \gamma xy - 30x - 60y + 225 = 0$, then $\alpha + \beta + \gamma$ is equal to

[JEE Mains-2025]

(A) 9

(B) 6

(C) 8

(D) 7

[Ans. : (A)]

DIBY-15

Let A and B be the two points of intersection of the line $y + 5 = 0$ and the mirror image of the parabola $y^2 = 4x$ with respect to the line $x + y + 4 = 0$. If d denotes the distance between A and B , and a denotes the area of $\triangle SAB$, where S is the focus of the parabola $y^2 = 4x$, then the value of $(a + d)$ is

[JEE Mains-2025]

[Ans. : 14]

DIBY-16

Let $A(0, 1)$, $B(1, 1)$ and $C(1, 0)$ be the mid-points of the sides of a triangle with incentre at the point D . If the focus of the parabola $y^2 = 4ax$ passing through D is $(\alpha + \beta\sqrt{2}, 0)$, where α and β are rational numbers, then $\frac{\alpha}{\beta^2}$ is equal to

[JEE Mains-2023]

- (A) 6 (B) 8 (C) 12 (D) $\frac{9}{2}$

[Ans. : (B)]

DIBY-17

If the line $3x - 2y + 12 = 0$ intersects the parabola $4y = 3x^2$ at the points A and B , then at the vertex of the parabola, the line segment AB subtends an angle equal to

[JEE Mains-2025]

- (A) $\tan^{-1}\left(\frac{11}{9}\right)$ (B) $\tan^{-1}\left(\frac{9}{7}\right)$ (C) $\tan^{-1}\left(\frac{4}{5}\right)$ (D) $\frac{\pi}{2} - \tan^{-1}\left(\frac{3}{2}\right)$

[Ans. : (B)]

DIBY-18

Let $y^2 = 12x$ be the parabola and S be its focus. Let PQ be a focal chord of the parabola such that $(SP)(SQ) = \frac{147}{4}$. Let C be the circle described taking PQ as a diameter. If the equation of a circle C is $64x^2 + 64y^2 - \alpha x - 64\sqrt{3}y = \beta$, then $\beta - \alpha$ is equal to

[JEE Mains-2025]

[Ans. : 1328]

DIBY-19

Let the point, P of the focal chord PQ of the parabola $y^2 = 16x$ be $(1, -4)$. If the focus of the parabola divides the chord PQ in the ratio $m : n$, $\gcd(m, n) = 1$, then $m^2 + n^2$ is equal to
[JEE Mains-2025]

- (A) 17 (B) 10 (C) 37 (D) 26

[Ans. : (A)]

DIBY-20

Let the focal chord PQ of the parabola $y^2 = 4x$ make an angle of 60° with the positive x-axis, where P lies in the first quadrant. If the circle, whose one diameter is PS , S being the focus of the parabola, touches the y-axis at the point $(0, \alpha)$, then $5\alpha^2$ is equal to
[JEE Mains-2026]

- (A) 15 (B) 25 (C) 30 (D) 20

[Ans. : (A)]

DIBY-21

If the length of the latus rectum of a parabola, whose focus is (a, a) and the length at its vertex is $x + y = a$, is 16, then $|a|$ is equal to
[JEE Mains-2022]

- (A) $2\sqrt{2}$ (B) $2\sqrt{3}$ (C) $4\sqrt{2}$ (D) 4

[Ans. : (C)]

DIBY-22

The area (in sq. units) of an equilateral triangle inscribed in the parabola $y^2 = 8x$, with one of its vertices on the vertex of this parabola, is **[JEE Mains-2020]**

- (A) $64\sqrt{3}$ (B) $256\sqrt{3}$ (C) $128\sqrt{3}$ (D) $192\sqrt{3}$

[Ans.: (D)]

DIBY-23

If the lines $x + y = a$ and $x - y = b$ touch the curve $y = x^2 - 3x + 2$ at the points where the curve intersects the x-axis, then $\frac{a}{b}$ is equal to **[JEE Mains-2020]**

[Ans.: 0.5]

DIBY-24

The tangent to the parabola $y^2 = 4x$ at the point where it intersects the circle $x^2 + y^2 = 5$ in the first quadrant, passes through the point **[JEE Mains-2019]**

- (A) $\left(-\frac{1}{3}, \frac{4}{3}\right)$ (B) $\left(-\frac{1}{4}, \frac{1}{2}\right)$ (C) $\left(\frac{3}{4}, \frac{7}{4}\right)$ (D) $\left(\frac{1}{4}, \frac{3}{4}\right)$

[Ans.: (C)]

DIBY-25

The tangents at the points $A(1, 3)$ and $S(1, -1)$ on the parabola $y^2 - 2x - 2y = 1$ meet at the point P . Then the area (in unit²) of the triangle PAB is **[JEE Mains-2022]**

- (A) 4 (B) 6 (C) 7 (D) 8

[Ans. : (B)]

DIBY-26

The equation of a common tangent to the curves, $y^2 = 16x$ and $xy = -4$ is

[JEE Mains-2019]

- (A) $x + y + 4 = 0$ (B) $x - 2y + 16 = 0$ (C) $2x - y + 2 = 0$ (D) $x - y + 4 = 0$

[Ans. : (D)]

DIBY-27

Let A be a point on the x -axis. Common tangents are drawn from A to the curves $x^2 + y^2 = 8$ and $y^2 = 16x$. If one of these tangents touches the two curves at Q and R . Then $(QR)^2$ is equal to **[JEE Mains-2023]**

- (A) 64 (B) 76 (C) 81 (D) 72

[Ans. : (D)]

DIBY-28

The common tangents to the circle $x^2 + y^2 = 2$ and the parabola $y^2 = 8x$ touch the circle at the points P, Q and the parabola at the point R, S . Then the area of the quadrilateral $PQRS$ is [JEE (Adv.)-2014]

- (A) 3 (B) 6 (C) 9 (D) 15

[Ans. : (D)]

DIBY-29

Let A_1, B_1, C_1 be three points in the xy -plane. Suppose that the lines A_1C_1 and B_1C_1 are tangents to the curve $y^2 = 8x$ at A_1 and B_1 , respectively. If $O = (0, 0)$ and $C_1 = (-4, 0)$, then which of the following statements is (are) TRUE? [JEE (Adv.)-2024]

- (A) The length of the line segment OA_1 is $4\sqrt{3}$
 (B) The length of the line segment A_1B_1 is 16
 (C) The orthocentre of the triangle $A_1B_1C_1$ is $(0, 0)$
 (D) The orthocentre of the triangle $A_1B_1C_1$ is $(1, 0)$

[Ans. (A), (C)]

DIBY-30

The tangents at the points $A(1, 3)$ and $B(1, -1)$ on the parabola $y^2 - 2x - 2y = 1$ meet at the point P . Then the area (in unit²) of the triangle PAB is

- (A) 4 (B) 6 (C) 7 (D) 8

[Ans. : (D)]

DIBY-31

P and Q are two distinct points on the parabola $y^2 = 4x$ with parameters t and t_1 respectively. If the normal at P passes through Q , then the minimum value of t_1^2 is

[JEE Mains-2016]

- (A) 2 (B) 4 (C) 6 (D) 8

[Ans. : (D)]

DIBY-32

A triangle is formed by the tangents at the point $(2, 2)$ on the curves $y^2 = 2x$ and $x^2 + y^2 = 4x$, and the line $x + y + 2 = 0$. If r is the radius of its circumcircle, then r^2 is equal to

[JEE Mains-2023]

[Ans. : 10]



DIBY-33

If the point on the curve $y^2 = 6x$, nearest to the point $(3, 3/2)$ is (α, β) , then $2(\alpha + \beta)$ is equal to

[JEE Mains-2021]

[Ans. : 9]

DIBY-34

On the parabola $y = x^2$ the point least distance from the straight line $y = 2x - 4$ is

- (A) $(1, 1)$ (B) $(1, 0)$ (C) $(1, -1)$ (D) $(0, 0)$

[Ans. : (A)]

DIBY-35

If $P(h, k)$ be point on the parabola $x = 4y^2$, which is nearest to the point $Q(0, 33)$, then the distance of P from the directrix of the parabola $y^2 = 4(x + y)$ is equal to [JEE Mains-2023]

- (A) 2 (B) 4 (C) 8 (D) 6

[Ans. : (D)]

DIBY-36

If the x-intercept of a focal chord of the parabola $y^2 = 8x + 4y + 4$ is 3, then the length of this chord is equal to

[JEE Mains-2023]

[Ans. : 16]

DIBY-37

Let PQ be a chord of the parabola $y^2 = 12x$ and the midpoint of PQ be at $(4, 1)$. Then, which of the following points lies on the line passing through the points P and Q ?

[JEE Mains-2024]

- (A) $(3, -3)$ (B) $(2, -9)$ (C) $\left(\frac{3}{2}, -16\right)$ (D) $\left(\frac{1}{2}, -20\right)$

[Ans. : (D)]

DIBY-38

Let $P(\alpha, \beta)$ be a point on the parabola $y^2 = 4x$. If P also lies on the chord of the parabola $x^2 = 8y$ whose mid point is $\left(1, \frac{5}{4}\right)$. Then $(\alpha - 28)(\beta - 8)$ is equal to

[JEE Mains-2024]

[Ans. : 192]



DIBY-39

If $P(h, k)$ be a point on the parabola $x = 4y^2$, which is nearest to the point $Q(0, 33)$, then the distance of P from the directrix of the parabola $y^2 = 4(x + y)$ is equal to:

[JEE Mains-2023]

- (A) 8 (B) 2 (C) 6 (D) 4

[Ans.: (c)]

DIBY-40

If tangent & normal drawn at point P intersects axis of parabola at $(-1, 2)$ and $(3, 5)$, then focal radius of point P is

[Ans.: (5/2)]

DIBY-41

Tangents PA and PB are drawn from point P on the directrix to the parabola

$(x - 2)^2 + (y - 3)^2 = \frac{(5x - 12y + 3)^2}{169}$. Find the least radius of the circumcircle of the triangle PAB .

[Ans.: (46/13)]

DIBY-42

Let a parabola P be such that its vertex and focus lie on the positive x -axis at a distance 2 and 4 units from the origin, respectively. If tangents are drawn from $O(0,0)$ to the parabola P which meet P at S and R , then the area (in sq. units) of $\triangle SOR$ is equal to

[JEE Mains-2021]

- (A) 32 (B) $8\sqrt{2}$ (C) 16 (D) $16\sqrt{2}$

[Ans.: (C)]

DIBY-43

Let L_1 be a tangent to the parabola $y^2 = 4(x + 1)$ and L_2 be a tangent to the parabola $y^2 = 8(x + 2)$ such that L_1 and L_2 intersect at right angles. Then L_1 and L_2 meet on the straight line

[JEE Mains-2020]

- (A) $2x + 1 = 0$ (B) $x + 3 = 0$
(C) $x + 2y = 0$ (D) $x + 2 = 0$

[Ans.: (B)]

DIBY-44

Consider the parabola with vertex $\left(\frac{1}{2}, \frac{3}{4}\right)$ and the directrix $y = \frac{1}{2}$. Let P be the point where the parabola meets the line $x = -\frac{1}{2}$. If the normal to the parabola at P intersects the parabola again at the point Q , then $(PQ)^2$ is equal to [JEE Mains-2021]

(A) $\frac{15}{2}$

(B) $\frac{25}{2}$

(C) $\frac{125}{16}$

(D) $\frac{75}{8}$

[Ans.: (C)]



*It is never about **end result**
It's always about the **Journey***

THANKYOU

#future **IITians**